A Proposed Design of Conventional 4-Bit Carry Look-Ahead Adder Improving Performance

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***Abstract*— This paper presents a method towards the improved performance parameters of conventional CMOS based 4-bit carry look-ahead adder. Conventional CLA adder has high numbers of transistors and high input impedance due to which various performance aspects are affected. Due to high input impedance, its delay and power consumption are high. Therefore, to increase the performance and to reduce delay, we have proposed an advanced version of CLA adder where hybrid logic based XOR gate and GDI AND gates have been used as input to reduce the transistor count as well as to improve performance. Finally, performance of modified adder has been compared with the conventional CLA adder. We have noticed that modified CLA adder showed better performance than the conventional CLA adder. Simulation has been done with Cadence virtuoso 90nm technology.**

Keywords*— carry look ahead, hybrid logic, gate diffusion input, adder

I. INTRODUCTION

Efficient design architectures play a vital role in circuit design due to rapidly increasing demand of battery-powered devices where an optimal balance among speed, power and area consumption of processors is highly crucial [1-4]. Therefore, circuit designers tend to make tradeoff among circuit parameters in order to meet the design requirements of the system [5-7].

Adder circuits being a core element play a vital role in modern microprocessors [6-7]. Adders are encountered in critical paths of several arithmetic operations such as multiplication [8], binary bit comparison, subtraction etc. [9]. As a result, it is quite important to enhance the performance of adders. For this reason, adders should be designed within small area having fewer transistors. It is also known that delay and area of an adder are inversely related. Therefore, designers often face the challenge to improve the performance of adders expending less area. As a result, there are numerous research ongoing for implementing of new adder technologies based on various logic styles.

Deployment of fast logic circuits may act as a solution to improve the delay but logic gates can be only made faster to certain extends due to physical limitation of devices [10]. As a result, reduction of voltage and clock speed or application of faster gates won't be a solution to improve the performance. Hence, application of new design techniques will be a proper solution to improve the performance.

Many techniques of adders have already been invented by the researchers those have some advantages and limitations [11-12]. However, for high-speed systems, Full Adder based Ripple Carry Adder method cannot cope up with the speed requirement [13-14]. Therefore, Parallel Prefix Adders (PPA) is being widely used in order to improve the performance of adder circuit [15]. Simplicity and better performance of Carry Look Ahead Adder has made this design method quite popular among PPAs [16-17].

The existing architecture of CLA adder is solely made up of conventional CMOS logic (CCMOS logic) [18]. CCMOS logic has become quite popular due to several advantages of its design. Such advantages are:

- Simplicity of design process
- High robustness
- High drive power
- Versatility

To sweep away the disadvantages of conventional single logic design based CCMOS CLA architecture, a hybrid design process can bring fruitful results by decreasing the number of transistors [19-20]. Meanwhile, employment of hybrid style for modification of conventional CLA can make the circuit consume less power while improving speed [21-22].

This research aims to intensify and improve the performance of conventional 4-bit CLA with the help of hybrid design method. The proposed CLA employed hybrid gates in input side through which transistor count could be decreased. Meanwhile, the employed hybrid gates in this work have less power and delay in contrast to the existing counterparts. Thus, overall improvement in case of various performance aspects could be obtained.

We have organized this paper as follows. Section II includes discussion on conventional CLA adder design and section III discusses about the methodologies used for designing the proposed CLA adder. Presentation of simulation result is shown in section IV. In section V, result analysis and discussion between conventional CLA adder

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and the proposed Adder is presented. Concluding remarks is discussed in section VI.

II. Conventional Static CMOS CLA Adder

Circuit Diagrams of CLA terms (C_1,C_2,C_3,C_4) are shown [23]in the Fig . 1. The output of CLA adder is presented by the following equations [24].

$$C_{i+1} = G_i + P_i C_i$$

$$S_i = P_i \oplus C_i$$

Where the term i= 0,1,2,3. Here, G_i is called carry generate that represents AND operation between A_i and B_i .

$$G_i = A_i B_i$$

In addition, P_i is called carry propagate that is responsible for XOR operation between A_i and B_i.

$$P_i = A_i \oplus B_i$$

XOR and AND gates of Conventional CLA adder are presented in the Fig. 2. Here, AND gate has 6 transistors and XOR has 12. In CLA, it is not necessary to wait for previous block of carry terms to compute result because all the terms are generated at the same time. Thus, carry propagation delay is being reduced in the CLA circuit which is the main advantage of CLA circuit.

III. Proposed Design Methodology

We have modified the proposed CLA in the input side. XOR and AND gates are used as input of CLA adder. The outputs of carry bits (C4, C3, C2, and C1) and the XOR gates for generating outputs of Sum terms (S3, S2, S1, and S0) are kept same as the conventional design. That means, we have made changes to the input side whereas output was same as the existing design. The modified XOR and AND gates are shown in the Fig. 3.

A. Modified AND gate design for Generate Terms Gi

Gate diffusion technology (GDI) has been used for designing the modified AND gate. This technique helps to establish a circuit using small number of transistors. GDI circuits also have low power consumption [25]. The simplest GDI based AND gate has 2 transistors [26]. Therefore, voltage degradation acts as a major problem in this type of GDI based AND gate. Full swing output is very necessary in GDI based AND gate. Modified hybrid GDI AND gates from [27] does not associate with voltage degradation issue. This gate provides full swing output. For this reason, this gate is not responsible for voltage degradation. We have noticed another AND gate in [28] which requires 5 transistors. Nevertheless, we have used AND gate obtained from [27] because this AND gate has lower delay than AND gate in [28]. Thus, using AND in [27] as a replacement for AND gate in conventional design for generating G_i signals would speed up the circuit.

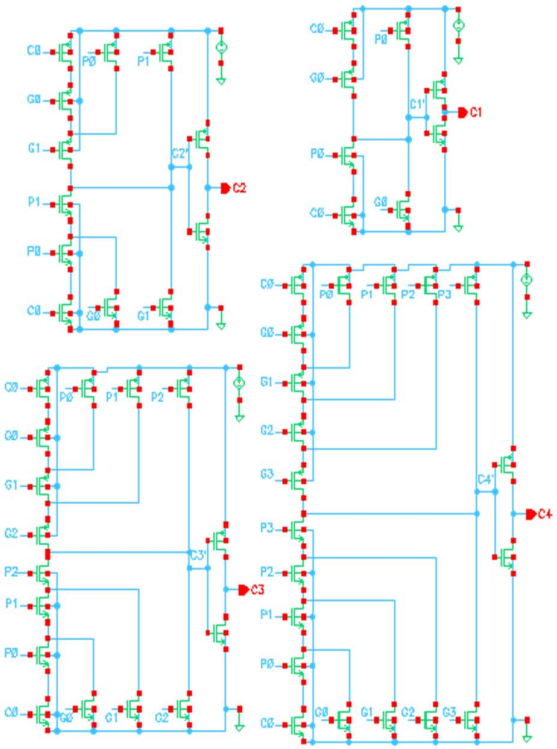


Fig. 1. CLA terms in conventinal S-CMOS logic

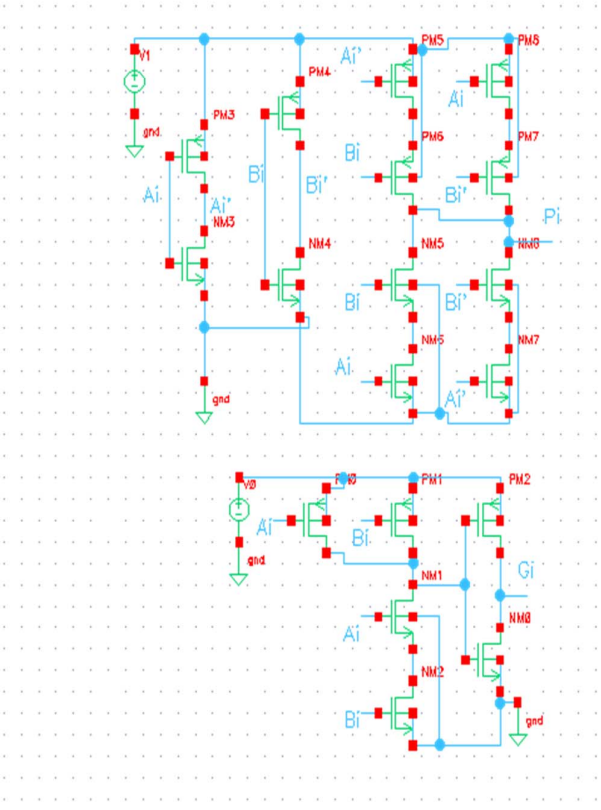


Fig. 2. XOR and AND gates for conventional design.

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B. *Modified XOR gate design for Propagate Terms Pi*

We have used modified hybrid XOR gate as a replacement of CCMOS XOR gate of existing design in the input side. Efficient hybrid logic style XOR gate having less transistors have been placed to the input side of CLA adder. The purpose is to reduce the input impedance. Besides, power consumption and circuit area will be less than previous for applying less number of transistor in this modified XOR gate. Therefore, the modified circuit has high driving power. For that we have used hybrid XOR gate to the input terminal of conventional CLA adder. This hybrid XOR gate has been taken from [29] as the XOR gate in this research performed better than others according to the simulation results. Thus, we have also got better performance for using this hybrid XOR gate.

IV. Simulation Result

We have done the simulation using Cadence Virtuoso Tools. For maintaining common environment for all the designs (proposed and existing), we have chosen 90nm technology and supply voltage has been set at 1.0 V. The simulated data on power, delay and power delay product (PDP) of the conventional designs and proposed design is presented on the Table. We have also presented the percentage of performance in this table.

We also listed the result in graphical manner so that it seems easier to differentiate the overall results. Average power consumption and delay of both adders are depicted in Fig. 4-5.

TABLE I. Transistor Count In Both Designs

CLA Adder Type	Total Transistors
Conventional Design	170
Proposed Design	138

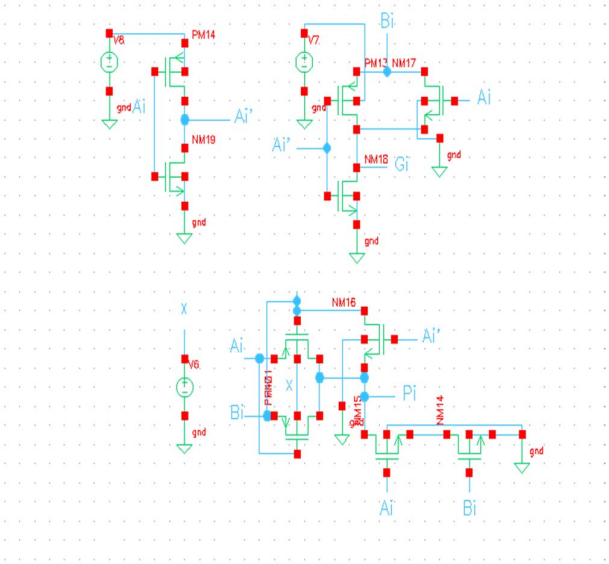


Fig. 3. XOR and AND gate used for proposed CLA

TABLE II. Simulation Output

CLA Adder Type	Power (µW)	Delay (ns)	PDP (fJ)
Conventional Design	55.39	0.163	9.03
Proposed Design	41.58	0.138	5.74
Improvement	24.93%	15.34%	36.43%

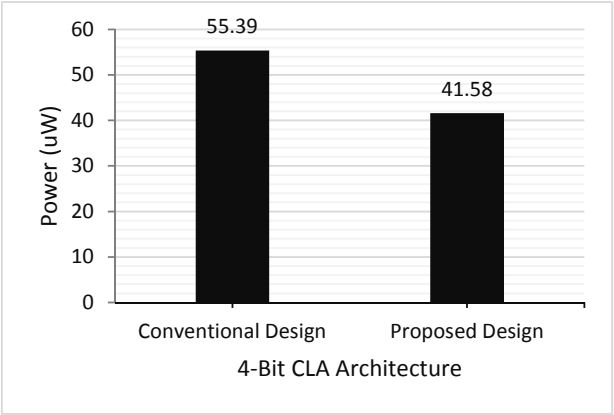


Fig. 4. Average power consumption of conventional and proposed design

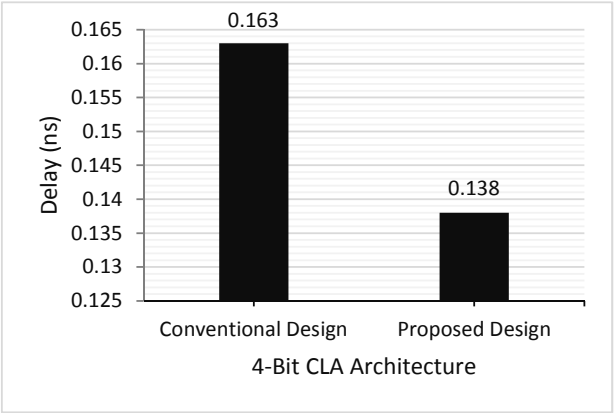


Fig. 5. Propagation delay of conventional and proposed design.

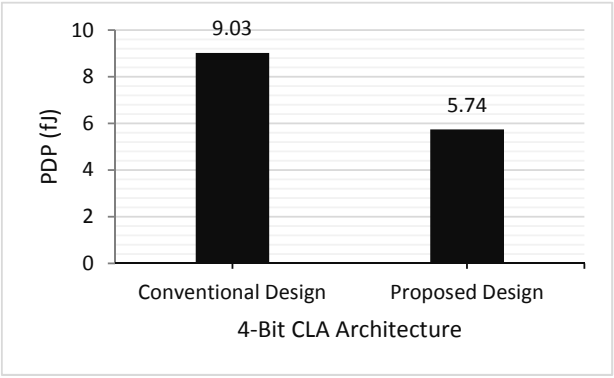


Fig. 6. Comparison of PDP between both adders

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V. Result Analysis and Discussion

We can notice from the result in Table I that the proposed CLA adder had excellent enhancement in overall power dissipation, speed and power delay product (PDP) than the conventional design. Moreover, the total transistor number in proposed design is less while comparing to the existing one. As a result, area and power of the proposed design is also lower than the conventional one which plays a vital role while designing CMOS circuits. The number of transistors in the conventional CLA adder is 170 whereas for proposed design transistor count is 138. Use of GDI gates in the input side decreased input impedance and it resulted in the reduction of delay. As GDI gates have been used only in the input side, it did not bring any changes in driving power of CLA adder.

Moreover, hybrid XOR gates have been used in the input side keeping advantages of conventional CLA adder in the proposed design. This is because driving power in output side is important to perform the circuit in high fan-out conditions. According to [6], it is better to utilize the advantages of different logic techniques rather than sticking to only one logic to get a better performance. Advantages of conventional design remain same in the modified design as the output terminals of both designs contain static CMOS logic. In addition, the modified XOR gate has less number of transistors which resulted in improving the overall performance. However, no modification made to the output side. This is because we have preserved the driving capability using Static CMOS logic. Thus, the proposed CLA adder reduced the drawbacks of existing design keeping its advantages.

VI. Conclusion

We have proposed a new design of CLA adder to reduce the transistor count and enhance its performance. To check the overall performance, we have done the simulation using Cadence technology. The proposed design showed better performance comparing with the conventional design. It showed 24.93 % improvement in average power and 15.34% improvement in delay. The improvement in PDP found 36.43% for changes in power and delay. Number of transistors in proposed design is also remarkably lower than conventional design. Besides, small number of transistors are responsible for less power consumption as well as less surface area. Therefore, the proposed adder design will also have lower surface area in comparison with the existing design. Thus, we can say that the proposed 4-bit CLA adder offers better performance than conventional CLA adder. Hence, use of proposed design in modern devices will certainly improve the performance of a system.

References

[1] A. Amouri, F. Arifin, F. Hannig and J. Teich, "FPGA implementation of an invasive computing architecture," 2009 International Conference on Field-Programmable Technology, Sydney, NSW, 2009, pp. 135-142.

[2] F. Arifin, R. Membarth, A. Amouri, F. Hannig and J. Teich, "FSM-controlled architectures for linear invasion," 2009 17th IFIP International Conference on Very Large Scale Integration (VLSI-SoC), Florianopolis, 2009, pp. 59-64.

[3] M. Hasan, M. H. Anik, S. Chowdhury, S. A. Chowdhury, T. I. Bilash, S. Islam, Low-Cost Appliance Switching Circuit for Microcontroller-Controlled Smart Home, International Journal of Sensors and Sensor Networks, vol. 2, no. 7, pp. 16-22, 2019.

[4] J. Warnock et al., "Circuit and Physical Design of the zEnterprise™ EC12 Microprocessor Chips and Multi-Chip Module," in IEEE Journal of Solid-State Circuits, vol. 49, no. 1, pp. 9-18, Jan. 2014.

[5] Y. Aizik and A. Kolodny, "Exploration of energy-delay tradeoffs in digital circuit design," 2008 IEEE 25th Convention of Electrical and Electronics Engineers in Israel, Eilat, 2008, pp. 1-5.

[6] M. Hasan, M. H. Anik and S. Islam, "Microcontroller Based Smart Home System with Enhanced Appliance Switching Capacity," 2018 Fifth HCT Information Technology Trends (ITT), Dubai, United Arab Emirates, 2018, pp. 364-367.

[7] M. Hasan, P. Biswas, M. S. Alam, H. U. Zaman, M. Hossain and S. Islam, "High Speed and Ultra Low Power Design of Carry-Out Bit of 4-Bit Carry Look-Ahead Adder," 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Kanpur, India, 2019, pp. 1-5.

[8] S. Gonzalez-Navarro, C. Tsen and M. J. Schulte, "Binary Integer Decimal-Based Floating-Point Multiplication," in IEEE Transactions on Computers, vol. 62, no. 7, pp. 1460-1466, July 2013.

[9] A. Monpapassorn, "Programmable wide range voltage adder/subtractor and its application as an encoder," in IEE Proceedings - Circuits, Devices and Systems, vol. 152, no. 6, pp. 697-702, 9 Dec. 2005.

[10] N. H. E. Weste, D. Harris, and A. Banerjee, CMOS VLSI Design: A Circuits and Systems Perspective, 3rd ed. Delhi, India: Pearson Education, 2010.

[11] M. Hasan, M. J. Hossein, U. K. Saha and M. S. Tarif, "Overview and Comparative Performance Analysis of Various Full Adder Cells in 90 nm Technology," 2018 4th International Conference on Computing Communication and Automation (ICCCA), Greater Noida, India, 2018, pp. 1-6.

[12] B. Harish, K. Sivani and M. S. S. Rukmini, "Performance comparison of various CMOS full adders," 2017 International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS), Chennai, 2017, pp. 3789-3792.

[13] A. Ibrahim, F. Gebali, "Optimized structure of hybrid ripple carry and hierarchical carry lookahead adders," Microelectronics Journal, vol. 46, no. 9, pp. 783-794, 2015.

[14] M. Hasan, U. K. Saha, A. Sorwar, M. A. Z. Dipto, M. S. Hossain and H. U. Zaman, "A Novel Hybrid Full Adder Based on Gate Diffusion Input Technique, Transmission Gate and Static CMOS Logic," 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Kanpur, India, 2019, pp. 1-6.

[15] K. C. Shilpa, M. Shwetha, B. C. Geetha, D. M. Lohitha, Navya and N. V. Pramod, "Performance Analysis of Parallel Prefix Adder for Datapath Vlsi Design," 2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT), Coimbatore, 2018, pp. 1552-1555.

[16] O. Akbari, M. Kamal, A. Afzali-Kusha and M. Pedram, "RAP-CLA: A Reconfigurable Approximate Carry Look-Ahead Adder," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 65, no. 8, pp. 1089-1093, Aug. 2018.

[17] S. Ghafari, M. Mousazadeh, A. Khoei, A. Dadashi, "A new high-speed and low area efficient pipelined 128-bit adder based on modified carry look-ahead merging with Han-Carlson tree method," 2019 MIXDES-26th International Conference on Mixed Design in Integrated Circuits and Systems, Rzeszow, Poland, June 2019, pp. 157-162.

[18] M. Binggeli, S. Denton, N. S. Muppaneni and S. Chiu, "Optimizing carry-lookahead logic through a comparison of PMOS and NMOS block inversions," 2015 IEEE International Conference on Electro/Information Technology (EIT), Dekalb, IL, 2015, pp. 641-646.

[19] R. F. Mirzaee, M. H. Moaiyeri, H. Khorsand and K. Navi, "A new robust and hybrid high-performance full adder cell", Journal of Circuits, Systems and Computers, vol. 20, no. 4, pp. 641-655, 2011.

[20] M. Hasan, M. J. Hossein, M. Hossain, H. U. Zaman and S. Islam, "Design of a Scalable Low-Power 1-bit Hybrid Full Adder for Fast Computation," in IEEE Transactions on Circuits and Systems II: Express Briefs, 2019.

[21] M. J. A. Morad, S. R. Talebiyan and E. Pakniyat, "Design of new low-power high-performance full adder with new XOR-XNOR

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circuit," 2015 International Congress on Technology, Communication and Knowledge (ICTCK), Mashhad, 2015, pp. 153-158.

[22] S. Lubaba, M. S. Islam, K. M. Faisal, M. Hasan, "Design of a two-bit magnitude comparator based on pass transistor, transmission gate and conventional static CMOS logic," 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Kharagpur, 2020, pp. 1-5.

[23] J. P. Uyemura, "Arithmetic Circuits in CMOS VLSI," in Introduction to VLSI Circuits and Systems, 2nd ed, New York, NY: Wiley, 2002, pp. 443-482.

[24] C. Efstathiou, Z. Owda and Y. Tsiatouhas, "New High-Speed Multioutput Carry Look-Ahead Adders," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 60, no. 10, pp. 667-671, Oct. 2013.

[25] M. Hasan, H. U. Zaman, M. Hossain, P. Biswas, S. Islam, "Gate Diffusion Input technique based full swing and scalable 1-bit hybrid full adder for high performance applications," Engineering Science and Technology, An International Journal, 2020.

[26] M. Hasan, U. K. Saha, M. S. Hossain, P. Biswas, M. J. Hossein and M. A. Z. Dipto, "Low Power Design of a Two Bit Mangitude Comparator for High Speed Operation," 2019 International Conference on Computer Communication and Informatics (ICCCI), Coimbatore, Tamil Nadu, India, 2019, pp. 1-4.

[27] M. Shoba and R. Nakkeeran, "GDI based full adder for energy efficient applications," Engineering Science and Technology, an International Journal, vol. 19, no. 1, pp. 485-496, 2015.

[28] M. Aguirre-Hernandez and M. Linares-Aranda, "CMOS full-adders for energy-efficient arithmetic applications," IEEE Trans. Very Large Scale Integr. (VLSI) Syst., vol. 19, no. 4, pp. 718–721, Apr. 2011.

[29] H. Naseri and S. Timarchi, "Low-Power and Fast Full Adder by Exploring New XOR and XNOR Gates," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 26, no. 8, pp. 1481-1493, Aug. 2018.

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